Pool ball Recognition Assignment

Computer Vision (CSU44053)

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1 Assignment Overview

The objective of this assignment was to develop a computer vision system capable of locating and recognising pool balls from photographs of a pool table. The two main tasks were to determine the position of each ball and to classify it as a solid, striped (along with each ball's repective number) or white cue ball.

The image dataset consisted of one reference image, AllBalls.jpeg, used for parameter tuning, and six test images, Table1–Table6.jpeg, containing different ball arrangements. All images were captured from the same camera viewpoint, and the four corners of the playing surface were provided.

The system was implemented in C++ using OpenCV, following a classical image-processing approach rather than a neural network. The implemented pipelines combines circular Hough detection, geometric filtering, perspective transformation, colour segmentation and rule-based classification.

2 System Design and Implementation

2.1 Pipeline Overview

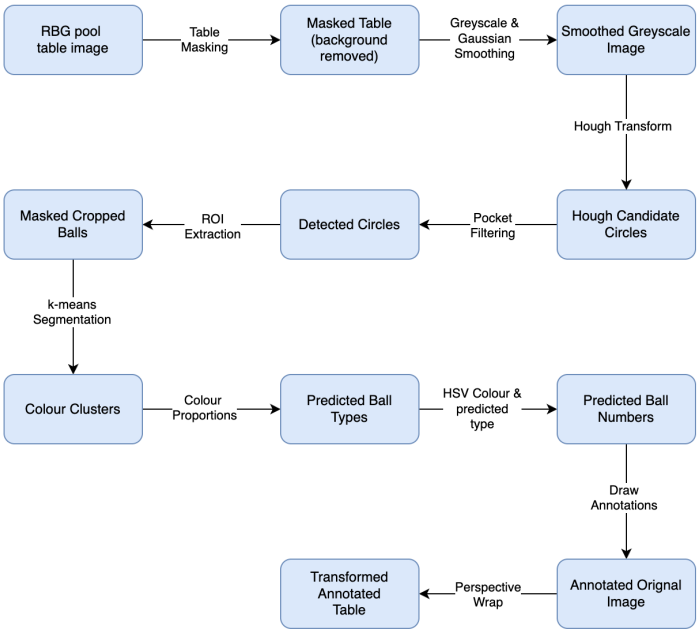


Figure 1: Overall processing pipeline, from input image to detected and classified balls.

The processing pipeline consists of two main stages: ball localisation and ball recognition. Localisation uses geometric and edge information to identify candidate circles. Ball recognition analysed the colour composition of each detected ball.

2.2 Ball Detection

Since the table boundaries were known and the camera position remained fixed, the playing surface could be represented by a quadrilateral. A binary mask was constructed from its four corners, restricting circle detection to the relevant image region and reducing interference from the surrounding environment. The goal of this step was to remove the background surrounding the pool table so that it would not interfere with ball detection.

The masked image was converted to grayscale and smoothed using a 5×5 Gaussian filter with a standard deviation of 1.5 (TODO: explain why these smoothing params). Smoothing was used as it reduces high-frequency noise and small local intensity variations that might otherwise introduce spurious edge responses.

Candidate ball locations were detected using OpenCV's circular Hough transform with the gradient-based method (`HOUGH_GRADIENT`). Applied to the grayscale, Gaussian-smoothed image, it identifies circular structures and returns their estimated centres and radii. These circles were treated as candidates because other features, such as pockets, could also produce detections.

2.2.1 Parameter tuning using grid search

The Hough transform requires several parameters that affect how many circles are detected and how accurately they match the balls. Rather than choosing these values only by visual inspection, a grid search was used to compare parameter combinations on the masked tuning image, `AllBalls.jpeg`.

The search varied four parameters:

- Accumulator resolution ratio: $dp \in \{1.0, 1.5, 2.0\}$.
- Minimum distance between detected centres: $\{15, 18, 20, 22, 25\}$ pixels.
- Upper Canny edge threshold: $\{75, 100, 125, 150\}$.
- Circle accumulator threshold: $\{15, 20, 25, 30, 35, 40, 45, 50\}$.

This produced 480 parameter combinations. The minimum and maximum circle radii were fixed at 15 and 30 pixels respectively.

For each combination, detected centres were transformed into table coordinates and matched to the annotated centres within a tolerance of 15 pixels. The best configuration was selected by first maximising the number of true positives, then minimising false positives, and finally minimising the mean centre localisation error.

The results for every combination were saved to a CSV file, together with an image showing the detections from the best configuration. Pocket filtering was not applied during this search; it was evaluated as a separate processing step.

2.2.2 Removing pocket detections

Table pockets can also produce circular detections. To remove these, candidate centres were transformed into table coordinates and compared with six approximate pocket positions. Candidates within 25 pixels of a pocket centre were discarded.

After the choosing the final Hough parameters and pocker filterting we obtained the following results on the tuning image:



Figure 2: AllBalls Ball detection with final Hough Params.

2.3 Ball Extraction and Colour Segmentation

Once the candidate balls have been detected, each ball is extracted from the original colour image using a square region centred on the predicted circle. A circular binary mask restricts subsequent processing to pixels belonging to the ball.

Initial experiments investigated colour thresholding, including HLS-based methods. However, a single global threshold was difficult to apply consistently across differently coloured balls, particularly in the presence of highlights, shadows and white markings.

The final implementation therefore uses K-means clustering to partition the colour pixels of each ball into two groups. Clustering is performed on BGR pixel values, and the resulting cluster centres are converted to HSV for interpretation.

The proportion of ball pixels assigned to each cluster is calculated. Each cluster's representative colour is then converted to HSV and categorised using the following rules:

- **Light:** brightness $V \geq 170$ and saturation $S \leq 165$.
- **Dark:** brightness $V \leq 90$.
- **Coloured:** all remaining clusters.

Here, saturation and brightness range from 0 to 255. The percentages of clusters in each category are added together to obtain the total light, dark and coloured proportions used for ball-type classification.

This approach reduces the original pixel-level information to a small set of interpretable features. It does not require manually defining a separate colour threshold for every individual ball.

2.4 Ball recognition

Ball recognition is performed in two stages. First, each detected ball is classified as white, solid or striped. Its predicted type is then combined with its colour to estimate its number.

2.4.1 Ball Type classification

Classification is performed using the proportions of light, dark and coloured pixels obtained from segmentation.

A ball is classified as white when at least 85% of its pixels belong to light clusters. It is classified as solid when at least 70% of its pixels belong to non-light clusters. Balls containing both light and non-light regions, without meeting the preceding criteria, are classified as striped.

These rules reflect the visual characteristics of pool balls: solid balls are predominantly coloured, while striped balls contain a substantial white region in addition to their coloured band.

However, this approach is sensitive to the visible orientation of the ball. A striped ball viewed from its white side may resemble the cue ball, while one viewed predominantly from its coloured side may resemble a solid ball.

An additional experimental classifier was developed using connected-component analysis. It examines the spatial organisation of light and non-light regions, including the largest connected components and

the proportion of the circular boundary occupied by light pixels. Its performance is compared with the simpler colour-based method in Section 3.

2.4.2 Ball number recognition

Number recognition uses the predicted ball type and the HSV colour of the largest eligible non-light cluster. Clusters covering less than 2% of the ball area are ignored, as are light clusters ($V \geq 170$ and $S \leq 165$).

A ball predicted as white is assigned number 0, representing the cue ball. For solid and striped balls, the selected cluster is examined using the following rules. OpenCV represents hue on a scale from 0 to 179, while saturation and brightness range from 0 to 255.

If $S < 60$, the ball is assigned number 8 only when $V \leq 90$ and its predicted type is solid. Otherwise, this low-saturation case returns an unknown result.

For clusters with $S \geq 60$, the colour thresholds in Table 1 determine the base number.

Table 1: Experimental HSV thresholds for ball-number recognition.

Colour	Base number	HSV condition
Yellow	1	$15 \leq H < 35$
Blue	2	$100 \leq H < 130$
Red	3	($H < 6$ or $H \geq 170$), $V \geq 150$
Purple	4	$130 \leq H < 165$
Orange	5	$6 \leq H < 15$
Green	6	$45 \leq H < 90$
Maroon	7	($H < 6$ or $H \geq 170$), $V < 150$

Solid balls retain the base number, while striped balls add 8, giving numbers 9–15. Red and maroon share the same hue range, so brightness is used to distinguish them.

When segmentation is unavailable, the predicted type is unknown, or no eligible cluster or colour match is found, the predicted number is set to -1. These thresholds are experimental and may be sensitive to changes in lighting.

2.5 Perspective Transformation

The table appears at an angle in the original photographs, while the provided ground-truth positions refer to a rectangular, top-down view. To compare our detections with these positions, their coordinates must be converted into the same coordinate system.

The four known table corners were mapped to the corners of a 1434×683 pixel rectangle. The resulting perspective transformation was then applied to each detected ball centre.

Ball detection and recognition were performed on the original image. Transforming the image beforehand could stretch the circular ball outlines into ellipses, making the Hough circle detector less reliable. The full image was transformed separately when producing visualisations for comparison with the ground truth.

TODO: Add image of the AllBalls image after perpective transformation

3 Evaluation and Results

3.1 Methodology

For the original detector and baseline evaluation, the reference image AllBalls.jpeg was used for parameter selection, while the six other images were reserved for evaluation. The later exploratory recognition search

in Section 3.4 also uses these six images for tuning; its results are reported separately. The original separation reduces the risk of reporting performance only on an image used to optimise the detector.

An automated grid search evaluated 480 combinations of Hough parameters. Configurations were ranked by the number of correctly detected balls, followed by the number of false positives and the average localisation error.

The final parameters were:

Table 2: Final Hough transform parameters for ball detection.

Parameter	Value	Description
`dp`	1.5	Image-to-accumulator resolution ratio
`minDist`	18	Minimum distance between centres (pixels)
`param1`	75	Upper Canny edge threshold
`param2`	35	Accumulator threshold for accepting circles
Minimum radius	15	Smallest accepted circle radius (pixels)
Maximum radius	30	Largest accepted circle radius (pixels)

Detected centres were compared with the provided ground truth after perspective transformation. A one-to-one matching procedure assigned detections to annotated centres within a maximum distance of 15 pixels in table coordinates.

Unmatched detections were counted as false positives (FP), while unmatched ground-truth balls were counted as false negatives (FN). Correctly matched detections were counted as true positives (TP).

Performance was assessed using precision, recall and F1-score. The mean Euclidean distance between matched centres was also calculated to evaluate localisation accuracy.

3.2 Ball Detection Results

On the tuning image, the selected configuration detected all 16 annotated balls, with three additional false detections. This gave a precision of 84.21%, recall of 100% and F1-score of 91.43%, before pocket filtering.

The subsequent evaluation on the six test images produced the following results using the full masked and pocket-filtered detection pipeline.

Table 3: Ball detection results on the validation images.

Image	GT	TP	FP	FN	F1
Table 1	16	14	2	2	0.8750
Table 2	16	16	2	0	0.9412
Table 3	16	15	0	1	0.9677
Table 4	15	14	1	1	0.9333
Table 5	14	14	1	0	0.9655
Table 6	13	13	1	0	0.9630
Total	**90**	**86**	**7**	**4**	**0.9399**

Across all six images, the detector achieved an overall precision of 92.47%, recall of 95.56% and F1-score of 93.99%. The mean localisation error across matched detections was approximately 5.01 pixels.

The relatively high recall indicates that the method successfully identifies most of the visible balls. However, its performance is less consistent when balls overlap or are closely grouped.

Table 1 produced the lowest detection F1-score and the largest mean centre error (7.68 pixels). This image contains a tightly arranged group of balls, where adjacent circular boundaries may interfere with Hough detection and accurate localisation.

By contrast, the later images contain more spatially separated balls, making individual circular outlines easier to identify. This is consistent with the generally stronger detection results on Tables 3–6.

3.3 Ball Recognition Results

Recognition was evaluated separately using only the detections successfully matched to ground-truth balls. This isolates classification performance from localisation errors.

The baseline K-means colour classifier correctly classified 76 of the 86 matched balls, giving an accuracy of 88.37%.

The experimental classifier incorporating connected-component features correctly classified 73 of the 86 matched balls, reducing accuracy to 84.88%.

Table 4: Ball recognition results for two different methods.

Method	Correct/Matched	Accuracy			
K-means colour rules	76/86	88.37%			
K-means + CCA + colour rules	73/86	88.37%			
Total	**90**	**86**	**7**	**4**	**0.9399**

The additional connected-component rules introduced three new classification errors without correcting any existing errors in this test set.

This suggests that spatial information is not necessarily more reliable than overall colour proportions. In particular, the shape and boundary coverage of white regions vary considerably with the orientation of striped balls. Consequently, rules intended to recognise stripes can also incorrectly modify predictions for the white cue ball or solid balls.

The simpler colour-based method was therefore retained in the main application.

The recognition accuracy of 88.37% is conditional on successful detection. Considering all 90 ground-truth balls, the system correctly detected and classified 76, corresponding to an end-to-end correct identification rate of 84.44%.

This distinction is important because a classification system cannot correctly identify a ball that was missed during localisation.

3.4 Exploratory Recognition Experiments

The baseline results above describe the original implementation. Two subsequent experiments investigated whether the extraction mask and recognition parameters could improve these results. Both experiments were implemented in a separate `experiments/` directory and executed using the independent `BallExperiments` target. They did not replace the classification functions, configuration values, or execution path of the main application. The results in this subsection therefore describe experimental alternatives rather than a change to the deployed baseline.

3.4.1 Evaluation protocol and interpretation

For these experiments, Table1–Table6 were used as an exploratory tuning set. AllBalls was reported separately as a reference image and was not included in the ranking objective of the recognition grid search. This differs from the earlier detection experiment, which selected Hough parameters on AllBalls and evaluated them on Table1–Table6. Since the recognition search used the labels of Table1–Table6 to select thresholds, its final scores on those images are tuning scores, not independent test estimates. Furthermore, all seven images share the same viewpoint and some unchanged ball positions. AllBalls is also the image on which the original rules were developed. Its separate results are consequently a useful regression check, but not evidence of generalisation to a new camera, table or lighting condition.

The detector, pocket filtering, perspective transformation and matching tolerance were held fixed. Detection was performed once per image. Predicted centres were transformed into table coordinates and matched one-to-one to annotated ball centres with a maximum distance of 15 table pixels, using the existing evaluation implementation. A match was established before varying recognition parameters and reused

for every configuration. The experimental run contained 90 annotated balls in Table1–Table6, of which 86 were matched. There were two unmatched candidate detections, which were excluded from recognition accuracy and displayed with an unavailable correctness indicator. All 16 annotated balls in AllBalls were matched. These are counts measured by the experiment's actual detection run; they are not inferred from a previously saved detection table.

For a matched ball, its annotated number determines its true type: number 0 denotes the white cue ball, numbers 1–8 are solid, and numbers 9–15 are striped. Candidate indices identify detections within an image and should not be interpreted as ball numbers.

Four measures were recorded. Let M be the number of matched detections, G the number of annotated balls, t_i the true type and n_i the true number of matched candidate i. Let $\hat{t}_i$ and $\hat{n}_i$ be the predicted type and number. The recognition accuracies are

$$A_{\text{type}} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}(\hat{t}_i = t_i), \qquad A_{\text{number}} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}(\hat{n}_i = n_i).$$

The number prediction uses the predicted type, so a type error can propagate into number recognition. An additional diagnostic reruns number classification with the true type supplied:

$$A_{\text{oracle}} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}\big(f_{\text{number}}(\text{clusters}_i, t_i) = n_i\big).$$

This diagnostic helps separate colour-selection failures from type errors. It is not an executable recognition solution because ground truth is required. In particular, supplying the true type of a cue ball invokes the existing number-0 shortcut when segmentation is available. Thus the measure should not be described as a pure hue-classification accuracy. Finally, end-to-end number recall divides correct detected-and-numbered balls by G instead of M. Missed balls remain errors under this measure. It does not measure the precision of numbered detections, since false-positive candidates are not included in its numerator or denominator.

3.4.2 Radius padding and cloth removal

The initial hypothesis was that Hough detection could underestimate a ball's radius and exclude useful pixels around its edge. The extraction radius was therefore changed from the rounded Hough radius r to $r' = r + p$, where $p \in \{0, 1, 2, 3, 4\}$ pixels. Both the square crop and its circular mask used r'. Enlarging the square alone would not admit additional pixels into colour analysis, because pixels outside the original mask would still be excluded. Crops were clipped to the image boundaries. Padding zero without filtering reused the original extraction function and reproduced the baseline mask.

Each padding value was evaluated with and without a local cloth-removal filter, producing ten variants per candidate. K was fixed at 2 and all type and number thresholds retained their original values. This isolated the effect of extraction from changes to the recognition rules.

The filter estimated the surrounding cloth colour from an annulus of width six pixels immediately outside the padded circle. Sampling used the original colour image. A sample was eligible when its OpenCV HSV values satisfied $35 \le H \le 95$, $S \ge 50$ and $V \ge 30$. These conditions restricted the estimate to visible green material rather than including arbitrary colours from nearby objects. If fewer than 20 eligible samples were available, filtering was skipped and the padded circular mask was retained.

The modal hue bin of eligible samples was selected. Saturation and brightness were averaged over samples within ten hue units of that mode. Pixels inside the candidate's mask were marked as potential cloth when their hue differed from the modal hue by at most 10, their saturation and brightness differed from the corresponding averages by at most 70, and they satisfied the same minimum saturation and brightness conditions. These tolerances use OpenCV's 0–179 hue and 0–255 saturation/brightness scales.

Potential cloth pixels were divided into eight-connected components. Only components touching the boundary of the circular mask were removed. The boundary was computed by subtracting a 3×3 elliptical erosion from the original mask, with pixels outside the crop treated as zero. This also recognised image-clipped edges as boundaries. Removed pixels were excluded from the mask rather than replaced by black pixels inside the analysis area. Consequently, K-means percentages used only the retained pixels as their denominator. The black background in diagnostic images was for display.

Boundary connectivity reduces the chance of deleting isolated green markings inside another ball, but does not distinguish a green ball from the cloth when their colours are similar and the matching region reaches the boundary. The observed masks showed that real green ball regions could be removed. Conversely, shadows and reflections could prevent some cloth pixels from matching the estimated colour. Cloth removal was therefore treated as an experimental option rather than assumed to be beneficial.

Table 5: Radius-padding experiment on matched balls in Table1–6 using the original recognition thresholds and K=2.

Padding (px)	Cloth filter	Matched	Correct type	Correct number
0	Off	86	76	67
1	Off	86	73	60
2	Off	86	67	49
3	Off	86	62	43
4	Off	86	60	42
0	On	86	77	69
1	On	86	76	66
2	On	86	78	64
3	On	86	72	55
4	On	86	66	50

Without filtering, increasing padding reduced correct number predictions on Table1–Table6 from 67 at zero padding to 60, 49, 43 and 42 at one through four pixels respectively. Cloth filtering partially compensated for the additional background, but padded variants still failed to improve number recognition over the original baseline. Two-pixel padding with filtering increased correct type predictions from 76 to 78, but reduced correct numbers from 67 to 64. The zero-padding filter alone gave 77 correct types and 69 correct numbers. On AllBalls, the unfiltered baseline already classified all 16 balls correctly; the filter at zero padding reduced both counts to 15.

These findings support retaining the original extraction radius for the next experiment. A likely explanation for the deterioration with padding is that additional cloth pixels alter cluster colours and proportions. This is an interpretation of the masks and recognition changes, not a direct measurement of the geometric accuracy of the Hough radius. The results do not establish that every detected radius equals the true visible ball boundary.

3.4.3 Recognition search: fixed processing and caching

The next experiment varied K, the interpretation of cluster colours and the type and number decision thresholds. Radius padding was fixed at zero. Cloth filtering remained an on/off factor, using the same filter parameters as the preceding experiment. Connected-component type correction was not included, so changes could be attributed to clustering and colour rules.

K-means was performed in three-dimensional BGR space using the existing implementation. The requested cluster counts were $K \in \{2, 3, 4, 5, 6\}$. For each call, the random-number generator state was set to 12345, three K-means++ initialisations were attempted, and the termination criteria were 50 iterations or an epsilon of 0.2. The prior random state was restored afterwards. These controls make comparisons repeatable within the same software environment; bit-for-bit equality across different OpenCV versions or platforms is not assumed.

Each candidate was segmented once for each K/filter combination. The run contained 104 candidates across all images, giving 1,040 cached segmentations. Only cluster percentages and representative BGR/HSV colours were retained for threshold evaluation. Cluster masks and segmented images were released after caching because subsequent rules depend on these features rather than on individual pixel labels. The original masked crops were retained for final diagnostic figures. No K-means rerun was required for each threshold configuration.

The experimental classifiers accepted explicit parameter structures rather than changing constants in the main classifier. Before searching, the experimental default type, percentages and number predictions were compared with the original production functions on every cached segmentation. Any disagreement terminated the experiment. This checked that baseline scores and parameterised scores represented the

same original decision rules.

3.4.4 Stage one: joint K and type-threshold grid

The first stage evaluated the full Cartesian product below. The original number thresholds were fixed during this stage, although changing the light thresholds also affected which cluster was eligible for number recognition. The same light definition was used in type classification and body-colour selection.

Table 6: Stage-one recognition search space.

Parameter	Evaluated values
Number of clusters K	2, 3, 4, 5, 6
Cloth filtering	Off, on
Minimum light-cluster V	150, 170, 190
Maximum light-cluster S	125, 145, 165, 185
Minimum light proportion for cue ball	80%, 85%, 90%
Minimum non-light proportion for solid ball	60%, 65%, 70%, 75%, 80%

This produced $5 \times 2 \times 3 \times 4 \times 3 \times 5 = 1{,}800$ configurations. A cluster was light when V met the minimum and S did not exceed the maximum. A ball was first considered for the cue-ball rule, then the solid rule, and finally the striped rule. Percentages and ordering therefore followed the original implementation. A failed segmentation or absence of sufficient evidence remained unknown and counted as incorrect for a matched ball.

The dark-cluster threshold was fixed at $V \leq 90$. Although it changes the separate reported dark and coloured percentages, it does not change a type decision: the solid and striped rules use the sum of these two categories. Searching it as a type parameter would duplicate equivalent predictions. The separate black-ball thresholds in number recognition were searched in stage two.

Configurations were ranked first by correct full-number predictions, then by correct type predictions, then by correct oracle-number predictions. All counts used Table1–Table6 only. Remaining ties favoured the smaller departure from the original parameters. The departure summed absolute threshold changes normalised by their search step sizes, plus $K - 2$ and one unit for enabling cloth filtering. The hue boundary at 165 used a five-unit normalisation even though its upper alternative was 168; the body-area threshold used absolute percentage-point change. Final ties favoured the earlier configuration ID. This tie rule prefers less change when measured performance is equal; it does not establish that the chosen thresholds are uniquely optimal. A separate best-type ranking reversed the first two accuracy criteria.

Table 7: Effect of K and cloth filtering on Table1–6. Default columns use the original thresholds. Tuned columns select the highest-ranked stage-one configuration within each K/filter pair; colour thresholds remain fixed.

K	Cloth	Default type	Default no.	Tuned type	Tuned no.	ID
2	Off	76/86	67/86	80/86	71/86	70
2	On	77/86	69/86	79/86	71/86	249
3	Off	73/86	61/86	81/86	69/86	370
3	On	77/86	67/86	80/86	70/86	610
4	Off	73/86	65/86	79/86	71/86	820
4	On	76/86	67/86	79/86	70/86	1060
5	Off	73/86	65/86	79/86	71/86	1179
5	On	76/86	68/86	79/86	71/86	1359
6	Off	74/86	66/86	81/86	73/86	1510
6	On	76/86	67/86	81/86	72/86	1750

The default-threshold columns show that increasing K alone did not improve number recognition. Without cloth filtering, the original K=2 configuration correctly numbered 67 balls, compared with 61, 65, 65 and 66 for K=3–6. However, after joint type-threshold tuning, K=6 without filtering achieved 73 correct numbers and 81 correct types. Thus the effect of K depended on how the resulting clusters were interpreted; the experiment did not support simply increasing K while leaving all thresholds fixed.

3.4.5 Stage two: grouped colour-threshold search

To limit the much larger colour search, a shortlist was formed from the top three distinct K/filter pairs under the full-number ranking and the top three distinct K/filter pairs under the type-first ranking. The original K=2 baseline, with and without cloth filtering, was also included. Exact duplicate seed configurations were removed. This run produced six seeds. Distinctness within each ranking avoided filling the shortlist with several nearly identical threshold settings for the same K/filter pair.

Each seed retained its K, cloth setting and stage-one type thresholds. Colour thresholds were searched in three consecutive groups:

1. Warm colours: red's upper hue boundary $\{4, 6, 8\}$; orange's upper boundary $\{12, 15, 18\}$; yellow's upper boundary $\{30, 35, 40\}$; red's wrap-around lower boundary $\{168, 170, 172\}$; and the red/maroon brightness split $\{130, 150, 170\}$. The nominal Cartesian product contained 243 combinations.
2. Green, blue and purple: green's lower boundary $\{40, 45, 50\}$; green's upper boundary $\{85, 90, 95\}$; blue's lower boundary $\{95, 100, 105\}$; the blue/purple boundary $\{125, 130, 135\}$; and purple's upper boundary $\{160, 165, 168\}$. This also gave 243 nominal combinations.
3. Neutral colours and body selection: black's saturation split $\{40, 60, 80\}$; black's maximum brightness $\{70, 90, 110\}$; and minimum eligible body-cluster area $\{1, 2, 5\}\%$. This gave 27 nominal combinations.

The intervals retained the original logic: red occupies both ends of the hue scale; orange and yellow follow its low-hue interval; green and blue have separate lower bounds; blue and purple share a boundary. Hue intervals must remain ordered, with strictly positive gaps between yellow and green, green and blue, and purple and high-hue red. Combinations violating these constraints were rejected. Existing unknown-colour gaps were therefore preserved rather than forcing every hue into a ball colour.

For each group, the current configuration was retained as a candidate. The group's best configuration under the same full-number ranking became the starting point for the next group. Two complete passes revisited warm, cool and neutral thresholds after later groups had changed. This was a sequence of exhaustive searches within small groups, not an exhaustive Cartesian search over every colour parameter jointly. It can miss a useful combination requiring simultaneous changes across groups, and it does not prove global optimality. Stage-one type thresholds were not revisited after colour tuning. Their remaining interactions with colour thresholds are another limitation of the staged search.

The actual second stage evaluated 5,526 valid combinations. Together with stage one this gave 7,326 scored trials. Trials repeated across groups, seeds or passes were retained in the CSV audit trail; this count is not the number of unique parameter vectors. Each trial recorded its parent configuration and stage so the selection sequence could be reconstructed. The minimum eligible cluster percentage was included for completeness, but with at most six clusters the largest non-light cluster often already exceeds all tested minimum areas. A parameter with no effect on these images should not be interpreted as precisely estimated.

3.4.6 Selected configuration and measured improvement

Table 8: Baseline and selected recognition parameters. Other colour parameters retain their original values.

Parameter	Baseline	Selected
K	2	6
Cloth filter (0=off)	0	0
Light minimum V	170	170
Light maximum S	165	125
Cue minimum light (%)	85	85
Solid minimum non-light (%)	70	80
Blue/purple hue boundary	130	125

Table 9: Exploratory recognition scores. Tuning uses Table1–6; the final column is full-number recognition on AllBalls, reported separately.

Configuration	Type	Type acc.	Number	Number acc.	Oracle	Ref. no.
Original baseline	76/86	88.37%	67/86	77.91%	70/86	16/16
Baseline + cloth filter	77/86	89.53%	69/86	80.23%	72/86	15/16
Best K/type grid	81/86	94.19%	73/86	84.88%	74/86	15/16
Best after colour search	81/86	94.19%	74/86	86.05%	75/86	16/16

The highest-ranked configuration used K=6, no cloth filtering, $V \geq 170$ and $S \leq 125$ for light clusters, an 85% light threshold for the cue ball, and an 80% non-light threshold for solids. Relative to the original rules, its blue/purple hue boundary moved from 130 to 125. Other colour thresholds remained unchanged. It was configuration 2153 in the saved run. The same configuration won both the full-number and type-first rankings after tie-breaking, so these names do not refer to two distinct selected models.

On the 86 matched tuning balls, type correctness increased from 76 to 81: 88.37% to 94.19%, a gain of 5.81 percentage points. Full-number correctness increased from 67 to 74: 77.91% to 86.05%, a gain of 8.14 percentage points. Five type errors and seven number errors were corrected relative to the baseline, with no previously correct predictions becoming incorrect in either category on these matched candidates.

The best stage-one configuration already produced 73 correct numbers. Colour tuning added one further correct number on Table1–Table6 and restored the stage-one reference score on AllBalls from 15/16 to 16/16. The latter restoration was observed after selection, not used to choose the hue boundary. Oracle-number correctness increased from 70/86 for the baseline to 75/86 for the selected configuration. It remained only one ball above full-number correctness, indicating that supplying the correct type alone resolved little of the remaining number-recognition error.

The cloth filter improved the original thresholds slightly, but was not part of the selected overall configuration. Its benefit was therefore conditional on K and the thresholds. These results support keeping cloth filtering optional rather than automatically adding it to the main pipeline.

Table 10: Per-image counts before and after the recognition search. All comparisons use the same matched candidates.

Image	GT	Matched	Base type	Best type	Base no.	Best no.
Table1.jpeg	16	14	8	11	6	9
Table2.jpeg	16	16	15	16	13	15
Table3.jpeg	16	15	14	14	14	14
Table4.jpeg	15	14	14	14	13	13
Table5.jpeg	14	14	14	14	11	12
Table6.jpeg	13	13	11	12	10	11

Improvements were unevenly distributed. On Table1, correct type predictions increased from 8/14 to 11/14 and correct numbers from 6/14 to 9/14. On Table2, correct numbers increased from 13/16 to 15/16. Tables5 and 6 each gained one correct number, while Tables3 and 4 retained their baseline number counts. This per-image view prevents the pooled gain from being mistaken for an equally large improvement on every scene.

Using all 90 annotations as the denominator, correct detected-and-numbered balls increased from 67/90 (74.44%) to 74/90 (82.22%). Detection itself was unchanged; the four missed balls could not be recovered by recognition tuning. The 16/16 reference result also matched, rather than exceeded, the original baseline on AllBalls.

Table 11: Selected configuration's type confusion matrix on Table1–6. Rows are ground truth and columns are predictions.

Actual / predicted	White	Solid	Striped	Unknown
White	6	0	0	0
Solid	0	42	1	0
Striped	3	1	33	0

The selected configuration still made five type errors and twelve full-number errors among matched tuning detections. For example, some striped balls were predicted as white because their visible light region dominated the crop. Some purple balls were assigned another colour or returned unknown; two black balls were assigned yellow. These examples show that tuning global thresholds does not resolve all effects of ball orientation, shadows, highlights and the selection of a single representative body cluster. They also explain why most remaining number errors persisted when the true type was supplied.



Figure 3: One matched candidate from Table1, compared under the original baseline, the cloth-filter baseline, the best stage-one configuration and the final selected configuration. Only final masked ball crops are shown. Each panel reports K, cloth setting, predicted type and number. Green text denotes a correct prediction and red denotes an incorrect prediction against the matched annotation. Identical winning configurations are displayed once.

3.4.7 Reproducibility, outputs and remaining limitations

The search is run from the repository root with `./build-local/BallExperiments--recognition-grid` after building the `BallExperiments` CMake target. Search ranges and original defaults are defined in `experiments/recognitionSearchConfig.h`. An optional output directory argument allows separate runs to be preserved.

Outputs are written under `output/experiments/recognition_grid_search/`. The type and colour leaderboards contain every evaluated configuration, its complete parameter vector, ranking metrics, stage and parent identifier. The K-comparison table records both the original thresholds and the best stage-one thresholds for each K/filter pair. Other CSV files contain the seed shortlist, finalist configurations, pooled and per-image metrics, candidate-level predictions, type and number confusion matrices, detection counts, mask sizes and cached cluster features. Unmatched candidates have blank correctness fields rather than being silently counted as correct or incorrect classifications. A separate image is saved for each candidate, with the baseline and distinct finalist configurations placed side by side.

Report tables and the diagnostic figure are local snapshots produced by `python3experiments/writeRecognitionReport.py`; the report can compile without the experiment outputs. Rerunning the search does not automatically update those snapshots. The surrounding interpretation must also be reviewed if parameters or measured results change.

The main limitation is selection on a small, correlated image set. Thousands of threshold trials can fit

properties specific to these balls, their orientations and the shared illumination. The measured gains should therefore motivate evaluation on fresh photographs before replacing the baseline. Neither the pooled tuning result nor the AllBalls reference check establishes performance on independent data. The search was also bounded to five K values, small threshold grids and two grouped colour passes; it establishes the best observed configuration under that procedure, not the best possible recognition system. No new production settings were applied as part of this experiment.

3.5 Qualitative Analysis and Failure Cases

Visual inspections revealed several recurring challenges.

Overlapping and touching balls. When balls are closely arranged, their visible boundaries may merge or be partially obscured. This can result in missed circles, imprecise centres or incorrect segmentation caused by pixels from neighbouring balls.

Ball orientation. A striped ball does not necessarily display equal amounts of white and colour. Depending on its orientation, the observed colour proportions may resemble either a solid ball or the white cue ball. This limitation is inherent to classification based on a single two-dimensional view.

Lighting and shadows. Highlights may cause coloured areas to appear white, while shadows can make coloured regions appear dark. Because classification depends on representative cluster colours and fixed HSV thresholds, these variations can change the predicted ball type.

Detection errors propagated into recognition. An inaccurate circle may include background pixels or exclude part of the ball. K-means then segments an imperfect crop, and the resulting percentages may no longer represent the actual ball.

These observations demonstrate why good localisation performance does not guarantee equally strong recognition performance.

4 Solution Anaysis and limitations

4.1 Limitations

Our method performs well on the supplied dataset, but several design choices limit its generalisation.

First, the fixed table mask and transformation depend on a known camera viewpoint. A different viewpoint would require new corner coordinates or an automatic table-localisation stage.

Second, a single set of Hough parameters may not be suitable for images containing balls with substantially different apparent sizes, lighting conditions or resolutions.

Third, two-cluster colour segmentation simplifies classification but may not adequately represent balls containing several visually distinct regions, such as the coloured band, white area, printed number and specular highlights.

4.2 Possible Improvements

Potential improvements include more robust ball-boundary extraction, adaptive colour classification and improved treatment of overlapping balls. More extensive testing across varied lighting conditions and viewpoints would also help assess generalisation.

Nevertheless, these extensions are not essential to the current assignment. The existing pipeline provides a complete solution with measurable performance and well-understood limitations.

5 Conclusion

This project implemented a classical computer vision pipeline for pool-ball localisation and recognition using geometric information, circular Hough detection, perspective transformation and colour-based segmentation.

The detector achieved an overall F1-score of 93.99% on the six test images, while the selected colour-based recognition method correctly classified 88.37% of successfully matched balls.

The results demonstrate that computer vision methods can produce strong localisation performance when the camera geometry is known. Recognition is more challenging because the visible colour distribution changes with ball orientation, illumination and occlusion.

The comparison with connected-component analysis also demonstrated that additional processing does not necessarily improve accuracy. Overall, the project achieved its two primary objectives while identifying concrete limitations and directions for future work.

6 Appendix

6.1 Use of Generative AI

6.2 Resources